

Path Smoothing via Footprint-Aware Corner-Transition Optimization for Differential-Drive Mobile Robots

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Abstract—ROS 2 Navigation 2 planners may produce abrupt heading changes that slow differential drive robots or require in place rotations. We propose PSTMO, a footprint aware smoother combining bounded waypoint removal, sampled swept footprint checks, curvature continuous quintic Bezier transitions, and differential drive feasibility tests. Dynamic programming selects a minimum cost nonoverlapping sequence of transitions and safe in place rotations. Tests on seven Gazebo map scenario pairs and five planners formed 35 groups. All PSTMO paths passed sampled static map checks. Across 34 complete groups, PSTMO reduced maximum translational curvature and mean squared curvature integral by 75.29 percent and 81.44 percent versus Simple, and by 80.62 percent and 92.10 percent versus Constrained. In closed loop tests, mean travel time was 6.96 percent below raw paths and 2.39 percent below Simple. Results indicate improved translational path smoothness and travel efficiency under the tested conditions.

Keywords—Differential-drive mobile robots, path smoothing, turning-maneuver optimization, quintic Bezier curves, swept-footprint collision checking, ROS 2 Navigation 2.